

Interactive Schwarzschild Black Hole Simulator

Project Planning Document – Natalia

1. Project Overview

This project is an **interactive scientific visualization application** that simulates the relativistic effects experienced near a **Schwarzschild black hole**.

The simulator will allow users to explore how gravity affects:

- the passage of time
- the frequency of light (gravitational redshift)
- the appearance of surrounding space through gravitational lensing

The core concept is based on a thought experiment frequently described in relativistic physics:

Two astronauts synchronize their clocks. One remains far from the black hole, while the other approaches the event horizon. Due to gravitational effects, the clocks begin to diverge.

The simulator visually demonstrates this phenomenon.

2. Primary Educational Goals

The application aims to demonstrate several key concepts from general relativity:

1. **Gravitational time dilation**
2. **Gravitational redshift**
3. **Event horizon behavior**
4. **Spaghettification near the horizon**
5. **Gravitational lensing of background stars**

The interface will also include an educational panel describing the physics involved.

3. Simulation Scenario

The simulation contains two observers:

Observer A – Distant Observer

- Remains stationary at a safe distance from the black hole.
- Serves as the **reference frame**.
- Clock advances at the global simulation rate.

Observer B – Interactive Astronaut

- Located in a spacecraft controlled by the user.
- Can move radially toward the black hole.
- Experiences relativistic effects.

The spacecraft can move between:

far region → strong gravity region → event horizon → interior

4. Physical Model

The black hole in this simulation is a **Schwarzschild black hole**, meaning:

- non-rotating
- uncharged
- perfectly spherical
- described by the Schwarzschild metric

This choice simplifies the physics while still allowing accurate representation of major relativistic effects.

5. Core Equations Used

Schwarzschild Radius

The radius of the event horizon is defined by:

$$r_{\square} = 2GM / c^2$$

Where:

- G = gravitational constant
- M = mass of the black hole
- c = speed of light

When the spacecraft reaches this radius, it crosses the event horizon.

Gravitational Time Dilation

The relationship between time measured far away and time measured near the black hole:

$$t_{\text{local}} = t_{\text{far}} \times \sqrt{1 - 2GM / rc^2}$$

Effects:

- As the spacecraft approaches the horizon, time slows relative to the distant observer.
 - At the event horizon, time appears frozen from the distant perspective.
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Gravitational Redshift

Light emitted near the black hole shifts toward longer wavelengths:

$$\lambda_{\text{observed}} = \lambda_{\text{emitted}} / \sqrt{1 - 2GM / rc^2}$$

Visual result in the simulator:

white → yellow → orange → deep red

6. Phenomena Simulated

The simulator represents several physical effects.

Time Dilation

The two astronaut clocks diverge as the spacecraft approaches the black hole.

The closer astronaut experiences slower local time.

Gravitational Redshift

Light emitted by the spacecraft shifts toward the red end of the spectrum.

The visual color of the spacecraft gradually changes based on its distance from the black hole.

Event Horizon

At the Schwarzschild radius:

- signals can no longer escape
- the spacecraft disappears from the external observer's universe

The simulator triggers a visual transition when this boundary is crossed.

Spaghettification

Near the horizon, tidal forces stretch objects dramatically.

The simulator approximates this effect visually:

- spacecraft elongates
- geometry distorts
- dotted outlines appear

This is a stylized visual effect rather than a full physical simulation.

Gravitational Lensing

The black hole bends the path of nearby light.

This produces visual effects such as:

- warped star fields
- distorted background geometry
- Einstein rings

This effect will be implemented through a **shader-based gravitational lensing distortion** applied to the starfield.

7. Simulation World

The simulated environment contains the following objects:

- Schwarzschild black hole
- accretion disk
- distant spacecraft (observer A)
- interactive spacecraft (observer B)
- starfield background

The black hole sits at the origin of the coordinate system.

(0,0,0) = black hole center

Distances are measured relative to the Schwarzschild radius.

8. Distance Scaling

The simulation does not use real meters.

Instead it uses **normalized units based on the Schwarzschild radius**.

Distance scale:

- 0 center of black hole
- 1 $r_{\square}$ event horizon
- 2 $r_{\square}$ extreme gravity
- 5 $r_{\square}$ strong gravity
- 20 $r_{\square}$ near flat spacetime

This normalization keeps numbers stable and understandable.

9. Simulation Architecture

The system is divided into several layers.

Application Layer

Controls simulation flow.

Responsibilities:

- start and stop simulation
 - manage simulation state
 - coordinate systems
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Domain Layer

Contains all scientific logic.

Components:

- physics engine
 - simulation engine
 - world state model
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Rendering Layer

Responsible for visual representation.

Components:

- 3D scene
 - black hole renderer
 - spacecraft renderer
 - gravitational lensing shader
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Interface Layer

Handles all user interactions.

Components:

- control panel
 - equation panel
 - camera controls
 - view switching
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Infrastructure Layer

Low-level systems:

- asset loading
 - timing system
 - event system
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10. Simulation Loop

Each simulation frame executes the following sequence:

1. read user input
2. update spacecraft distance
3. compute gravitational effects
4. update astronaut clocks
5. update visual parameters
6. render the scene

The physics simulation is independent from the rendering system.

11. Camera System

The simulator uses a **3D orbital camera**.

Users can rotate around the black hole to observe:

- the spacecraft trajectory
- the accretion disk
- the lensing of background stars

Camera limits prevent clipping through the event horizon.

12. Simulation Views

The interface will support three viewing modes.

External Observer View

Shows the entire system:

- black hole
- spacecraft

- starfield
 - accretion disk
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Distant Astronaut View

Perspective from the stationary spacecraft.

Effects:

- approaching spacecraft appears to slow
 - extreme redshift near horizon
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Near Astronaut View

Perspective from the interactive spacecraft.

Effects:

- outside universe appears accelerated
 - gravitational distortions increase
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13. User Interaction

Users can control the spacecraft using:

- radial movement controls
- distance slider
- drag interaction

Movement range:

far region → center of black hole

Once the horizon is crossed, normal control stops.

14. Scientific Panel

The simulator includes a lateral panel explaining the physics.

Sections:

Theory

Explanation of Schwarzschild black holes.

Equations

Displays the formulas used in the simulation.

Live Variables

Displays real-time data:

- distance from black hole
- Schwarzschild radius
- time dilation factor
- redshift factor
- astronaut clock values

Controls

Simulation settings.

15. Visual Design

The visual style will include custom artwork created by the developer.

Visual elements include:

- stylized spacecraft models
- animated accretion disk
- gravitational lensing distortion
- starfield background

Performance will be prioritized over excessive geometry.

16. Performance Constraints

Target performance:

60 FPS

Design limits:

≤ 20k polygons
≤ 1024px textures
minimal shaders

These limits ensure the simulator runs smoothly in a browser.

17. Splash Screen

The application begins with a short introduction screen.

Contents:

- developer logo
- developer name
- simulator title

Duration:

2–3 seconds

The splash transitions smoothly to the simulation scene.

18. Development Phases

The project will be developed in sequential stages.

Phase 1

Physics engine

Phase 2

Simulation state system

Phase 3

Clock synchronization and divergence

Phase 4

Basic 3D scene

Phase 5

Spacecraft interaction

Phase 6

Relativistic visual effects

Phase 7

Gravitational lensing shader

Phase 8

User interface and panels

Phase 9

Optimization and testing

19. Portfolio Presentation

The project will include documentation explaining:

- system architecture
- relativistic physics model
- rendering techniques
- screenshots
- demonstration video

The final result will be presented as an **interactive educational physics simulator**.

20. Project Title

Working title:

Interactive Schwarzschild Black Hole Simulator

Created by Natalia.